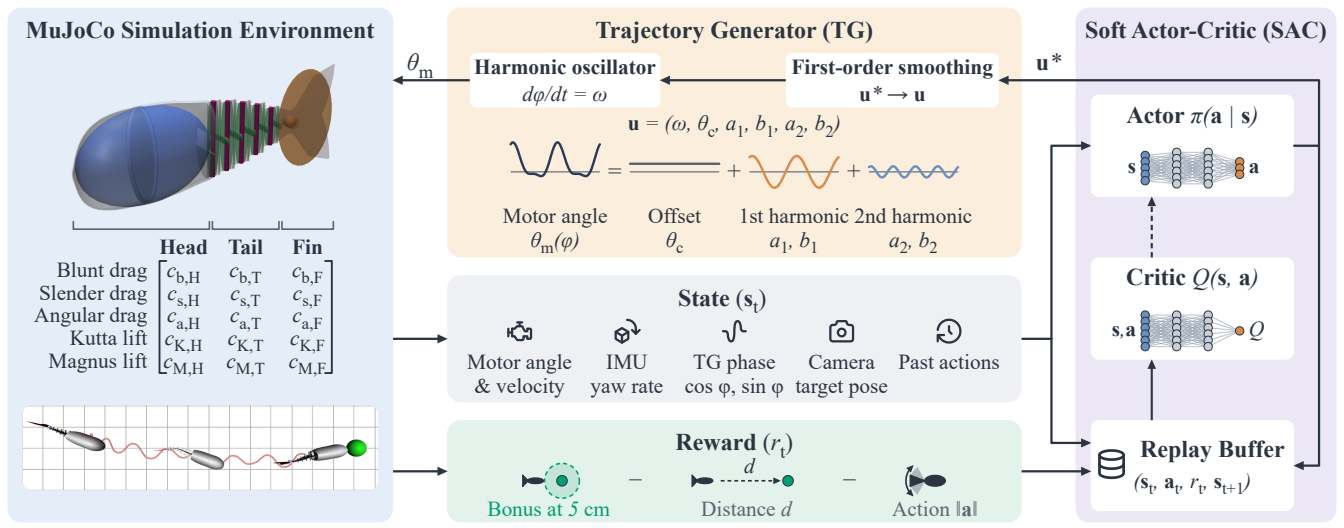

MuJoCo Simulation Environment
Head
Tail
Fin
Blunt drag
Slender drag
Angular drag
Kutta lift
Magnus lift
$c_{b,H}$ $c_{b,T}$ $c_{b,F}$
$c_{s,H}$ $c_{s,T}$ $c_{s,F}$
$c_{a,H}$ $c_{a,T}$ $c_{a,F}$
$c_{K,H}$ $c_{K,T}$ $c_{K,F}$
$c_{M,H}$ $c_{M,T}$ $c_{M,F}$
Trajectory Generator (TG)
θ_m
Harmonic oscillator
$d\varphi/dt = \omega$
First-order smoothing
$\mathbf{u}^* \rightarrow \mathbf{u}$
$\mathbf{u}^*$
$\mathbf{u} = (\omega, \theta_c, a_1, b_1, a_2, b_2)$
Motor angle
$\theta_m(\varphi)$
Offset
θ_c
1st harmonic
a_1, b_1
2nd harmonic
a_2, b_2
Soft Actor-Critic (SAC)
Actor $\pi(\mathbf{a} \mid \mathbf{s})$
s
a
Critic $Q(\mathbf{s}, \mathbf{a})$
s,a
Q
State ($\mathbf{s}_t$)
Motor angle & velocity
IMU yaw rate
TG phase cos φ, sin φ
Camera target pose
Past actions
Reward (r_t)
Bonus at 5 cm
d
Distance d
Action $\|\mathbf{a}\|$
Replay Buffer
$(\mathbf{s}_t, \mathbf{a}_t, r_t, \mathbf{s}_{t+1})$